

BlockPulse Crypto Market Intelligence

Automated Data Pipeline & Analytics for Cryptocurrency Markets

Objectives & Use Case

Goal

Capture daily cryptocurrency market snapshots, transform raw data into a star schema and make it query-ready for analytics and dashboards.

Motivation

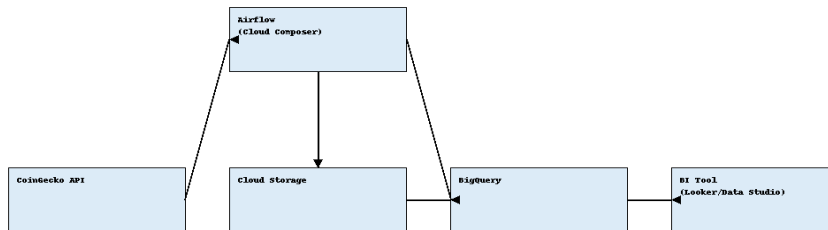
Crypto markets are fast-moving and fragmented. Manual data gathering is time-consuming and error-prone. A robust pipeline ensures data quality and frees analysts to focus on insight.

- Top-N coins captured daily
- Clean, star-schema warehouse
- Cloud-native and scalable
- Ready for dashboards & AI

Outcome

A curated warehouse powering dashboards that track prices, market cap, trading volumes and rankings for the top cryptocurrencies.

Pipeline Architecture



Sources

CoinGecko API provides price, market cap and volume metrics.

Orchestration

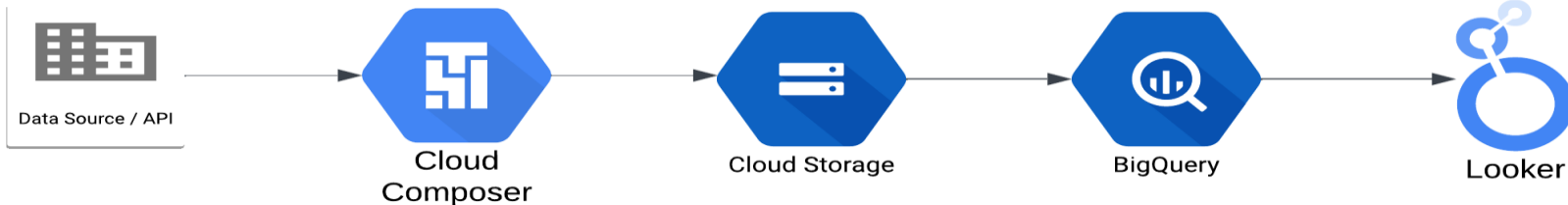
Airflow (Cloud Composer) coordinates extraction, loading and transformation.

Storage & Processing

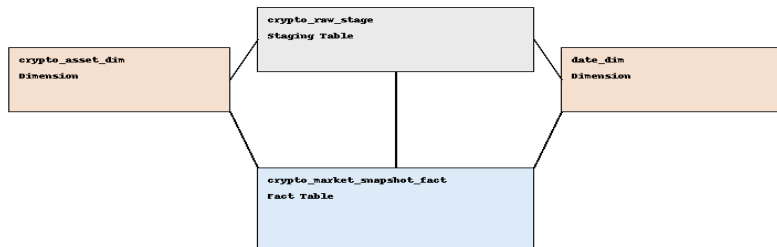
Raw data lands in GCS; BigQuery stores staging, dimension and fact tables.

Analytics

Looker/Data Studio dashboards consume the warehouse for interactive insights.



Dimensional Model



Fact Table

crypto_market_snapshot_fact stores daily metrics per coin (price, volume, market cap, rank, change).

Dimension Tables

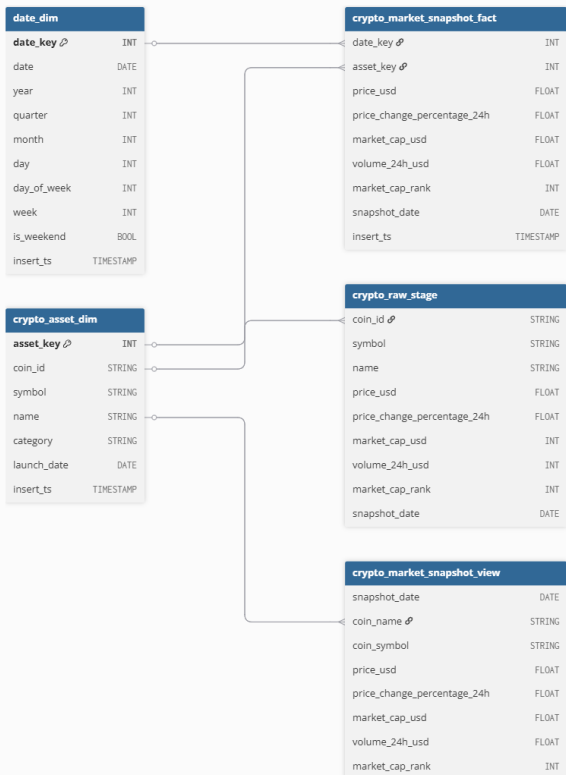
crypto_asset_dim holds metadata for each coin (symbol, name, category, launch date).

date_dim provides calendar attributes for each snapshot date (year, quarter, month, day of week).

Staging Table

crypto_raw_stage acts as a landing zone before transformation and upserts.

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Implementation & Tools

Python Scripts

fetch_api_data: call CoinGecko, normalise JSON and write CSV.

load_to_gcs: upload CSV snapshot to Google Cloud Storage.

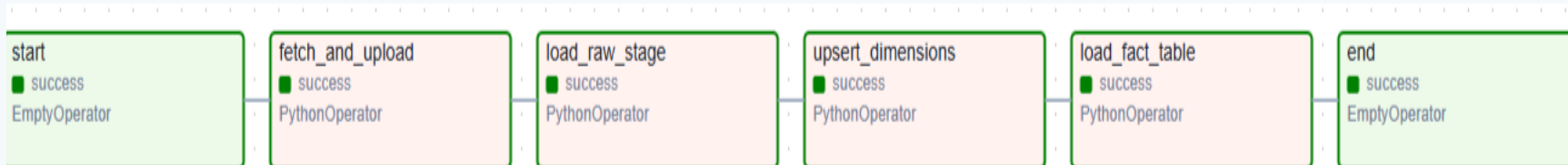
load_to_bq: load CSV into BigQuery (raw stage) with schema autodetect.

Airflow DAG

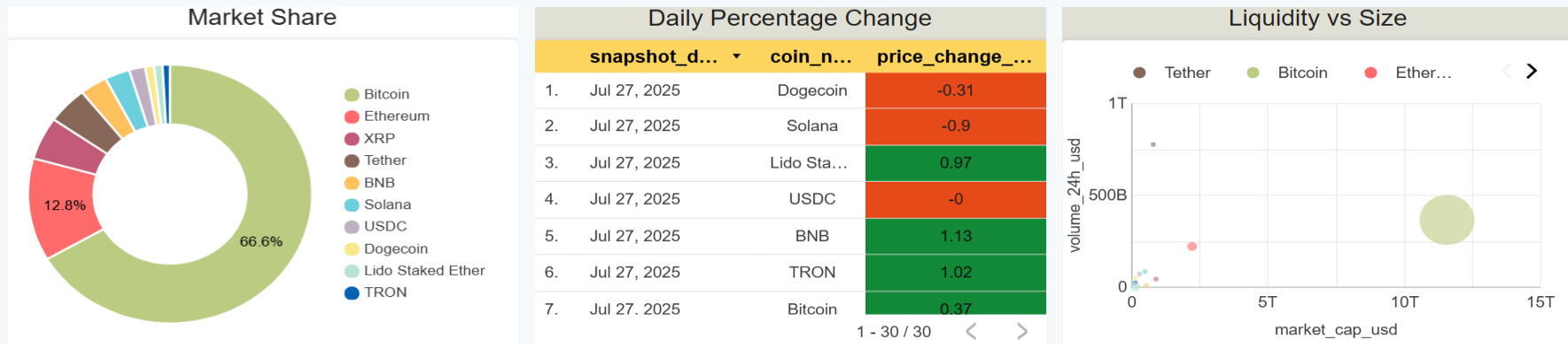
Defines a daily workflow: start → fetch/upload → load stage → upsert dimensions → load fact → end.

SQL Procedures

Stored procedures upsert new assets into crypto_asset_dim and new dates into date_dim, then insert records into the fact table.



Dashboard Design



Three Sections

1. Market Snapshot: stacked area and bar charts plus market-share donut to show overall trends.
2. Coin Performance: multi-series line chart, rank bump chart and a heatmap for daily changes.
3. Comparative Insights: bar charts and bubble plot to compare size and liquidity across coins.

Interactive filters for date range and coin selection ensure the narrative adapts to the user's interests.

Links

Dashboard

<https://lookerstudio.google.com/s/mcn2g1qssxA>

GitHub Repo

<https://github.com/euphoricking/blockpulse>

Optimisation & Future Work

Scale & Flexibility

Parameterise the list of coins and currencies, add pagination to API calls and parallelise tasks for faster loads.

Cost Optimisation

Partition and cluster BigQuery tables on `snapshot_date` and `asset_key`, compress data (gzip) in GCS and purge old raw snapshots.

Monitoring & Alerts

Enable Cloud Monitoring for Airflow, set up failure alerts and track load job statistics. Use dashboard-based health checks to catch anomalies early.

Extensions

Integrate additional metrics (e.g. volatility, sentiment), add more dimensions (e.g. exchange, region) and experiment with real-time streaming ingestion.

Conclusion

BlockPulse Insights delivers a resilient, scalable pipeline that turns raw crypto market data into actionable intelligence.

A star-schema warehouse powers flexible dashboards for traders, analysts and researchers.

By automating ingestion and embracing best-practices in cloud engineering, the platform frees you to focus on insight rather than plumbing.

Thank you! Questions?